

TSFA: A Two-Stage Feature Alignment Method for Unsupervised Open-Set Domain Adaptation in Time-Series Classification

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Abstract—The unsupervised open-set domain adaptation (UOSDA) in computer vision is widely studied. However, designing UOSDA algorithms for time series remains challenging due to the complex nonstationary property of data and distribution shifts across different operating conditions, which heighten the risk of negative transfer. To address this, a novel two-stage feature alignment (TSFA) method for UOSDA in time-series classification is proposed. First, a time-frequency feature extractor is designed to effectively learn domain-invariant and discriminative interclass representations. Subsequently, a two-stage multigranularity feature alignment framework is proposed. Specifically, global alignment is achieved by minimizing the similarity distribution entropy (SDE), which encourages similar samples from both domains to converge in the feature space, thereby reducing intraclass distances of all target samples. Then, local alignment is performed via self-supervised learning guided by target pseudolabels, enhancing interclass discriminability of target common samples. Furthermore, an optimal similarity assignment matrix (OSAM) is designed to obtain more accurate pseudolabels for common samples, while an adaptive decision boundary is introduced to effectively reject private samples based on target-domain statistics. Finally, extensive experimental results on three real-world datasets indicate that the proposed method outperforms state-of-the-art approaches.

Index Terms—Domain adaptation, open set, optimal transport (OT), self-supervised, time series.

I. INTRODUCTION

NEURAL network and learning systems often encounter the challenges of different distributions between train and test data, such as images captured under varying weather or lighting conditions [1], [2]. Moreover, in the real world, private classes that are not present in the train datasets may occur [3]. Unsupervised open-set domain adaptation (UOSDA) overcomes these challenges by transferring knowledge from

the source domain to the target domain, improving the accuracy of common-class classification and the effectiveness of private-class identification in the target domain [4].

Notable theoretical works and effective UOSDA algorithms have been developed for visual applications. A pioneering $\mathcal{H}\Delta\mathcal{H}$ divergence was introduced to measure distribution shifts, and the strict generalization bounds were provided in [5]. Subsequently, various techniques were applied in domain alignment, including methods based on statistical distance [6], adversarial learning [7], self-ensembling [8], and pseudolabeling [9]. Methods such as setting a confidence threshold [10] and adding an additional classification head in the last layer of the classifier [11] are used to identify the private classes in the target domain.

Despite the limited number of studies, research on UOSDA for time-series data remains highly significant. In industrial applications, the continuous operation of equipment generates vast amounts of unlabeled time-series data. This presents critical challenges, including poor model generalization, strong dependence on labeled data, and difficulty in identifying previously unseen or unknown states [12], [13], [14]. Addressing these challenges requires transfer learning methods that generalize across devices, conditions, and scenarios.

Despite the significance of UOSDA for time-series classification, several challenges remain, as depicted in Fig. 1.

- 1) *Nonstationary Characteristics*: Time series often exhibit evolving statistical properties, leading to changes in both temporal and spatial features. This hides the extraction of cross-domain invariant features [15], [16], [17], [18].
- 2) *Negative Transfer*: As shown in Fig. 2, cross-domain samples from different classes exhibit large discrepancies in frequency features. Such gaps cause abnormal gradients in discrepancy-based alignment (e.g., KL or MMD), which in turn reduce class separability and induce negative transfer [18], [19].
- 3) *Pseudolabels*: Due to the nonstationary nature and complexity of time-series data, domain shifts can significantly increase the error rate of pseudolabels, especially when unknown classes are present [17].

To address the above challenges, a two-stage multigranularity feature alignment framework is proposed. The distinctions between two-stage feature alignment (TSFA) and existing methods are explicitly designed to tackle the three aforementioned challenges illustrated in Fig. 1. First, to address

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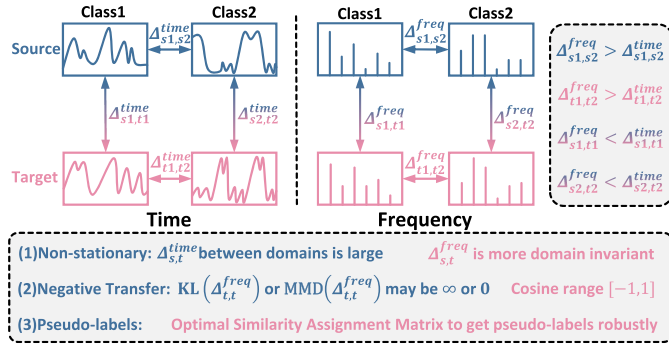


Fig. 1. Frequency features are more domain-invariant than time. Δ is the divergence between two time series, where the Euclidean distance is employed and subsequently normalized in the figure.

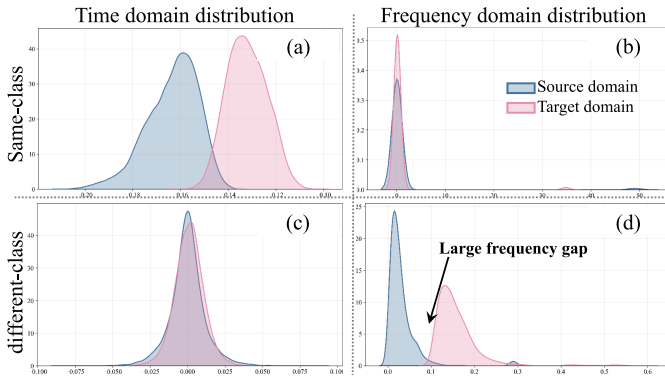


Fig. 2. Frequency features are more domain-invariant under KDE [20]. Comparison between (a) and (b) shows that the frequency features of the same class remain consistent across domains, while comparison between (c) and (d) demonstrates that the frequency features of different classes vary significantly across domains.

nonstationarity, we leverage the empirical observation that frequency features exhibit stronger domain invariance than temporal features (i.e., $\Delta_{s,t}^{freq} < \Delta_{s,t}^{time}$), as illustrated in Fig. 2. Motivated by this, TSFA incorporates a learnable weighting mechanism to adaptively select transferable frequency components, thereby extracting robust domain-invariant representations. Second, to mitigate negative transfer, **different from** conventional methods that directly align features based on discrepancies [18], [21], the SDE is designed. By normalizing features onto a hypersphere and optimizing **cosine similarity**, SDE effectively mitigates the gradient anomalies caused by large frequency discrepancies. Minimizing SDE in the feature space implicitly aligns target common samples with source prototypes and facilitates the self-clustering of private samples. Third, to rectify pseudolabel noise, the approach moves beyond local, threshold-based assignment. The optimal similarity assignment matrix (OSAM) is proposed, which formulates the pseudolabeling as a global assignment problem. **Different from** previous works that use optimal transport (OT) merely as a discrepancy loss function [22], [23], [24], OSAM leverages the global transport plan to generate accurate pseudolabels for common samples while adaptively distinguishing private samples. By accounting for global distribution constraints, OSAM provides a theoretically grounded and robust alternative to self-supervised alignment.

The novel contributions of this article can be summarized as follows.

- 1) A TSFA method based on a time–frequency feature extractor is proposed for UOSDA in time-series classification. By explicitly capturing both spatiotemporal and frequency characteristics, the extractor effectively learns domain-invariant representations, ensuring robustness against distribution shifts arising from diverse operating conditions.
- 2) A multigranularity feature alignment strategy is designed to enhance cross-domain interclass separability and intraclass consistency. The cosine similarity-based SDE component addresses the gradient instability caused by frequency features, while the local alignment module aligns precisely target common-class samples in a class-wise manner. Together, these components effectively mitigate negative transfer.
- 3) A pseudolabeling strategy based on the OSAM is proposed, which leverages the distributional information of an entire batch to achieve greater accuracy and stability. Furthermore, a domain-structure-adaptive thresholding mechanism is introduced. Unlike conventional fixed-threshold methods, this mechanism provides a robust rejection boundary for identifying private samples, adapting dynamically to the distinct statistics of the target domain.

The rest of this article is organized as follows. Section II reviews the related works. The proposed TSFA is elaborated in Section III. Section IV shows the extensive experimental results as well as the insightful analysis. The conclusion is discussed in Section V.

II. RELATED WORKS

In this section, a brief review of the recent UOSDA algorithms and domain adaptation methods for time series is conducted.

A. Unsupervised Open-Set Domain Adaptation

In UOSDA, models must not only address distribution discrepancies between the source and target domains but also identify new categories that appear in the target domain [25].

Modest theoretical attempts about UOSDA were conducted. For instance, the theoretical upper bound on the target-domain risk was established, and an open-set difference was introduced to enable the identification of novel classes [26], [27]. OVANet proposed a one-versus-all classifier for each category to learn source interclass distances and recognize private target samples [11]. The sample affinities used effectively distinguish common and private samples in the target domain [28]. A causality-driven UOSDA method was presented, which incorporates front-door adjustment (FDA) and decoupled causal alignment (DCA) for unbiased transfer [29]. STUN addressed UniDA by leveraging stochastic binary classifiers for robust confidence estimation, aligned common classes via weighted adversarial learning, and used deep discriminative clustering to address fragmented target-domain distributions [30]. GLC++ [31] integrated global–local clustering with

contrastive affinity learning for source-free UniDA. Iterative contrastive clustering [32] formulated open-world DA as an iterative process combining semisupervised clustering and contrastive learning. COSDA [33] proposed a counterfactual-based susceptibility risk framework with contrastive feature alignment and virtual multiunknown-category prototypes. An unknown exploration strategy was proposed, which exploited high-confidence negatives and synthetic adversarial samples to enhance the separation of unseen categories [34].

However, current transfer learning paradigms exhibit inherent limitations, such as the poor performance of MMD when distributions undergo dynamic changes [35] and the risk of class-specific information loss in adversarial learning [36]. Unlike GLC++ [31] and COSDA [33], which rely on standard features, TSFA integrates both time and frequency representations. This is crucial because frequency features are more domain-invariant for time series, offering a stable anchor that general methods miss. Compared to contrastive learning, SDE avoids the disruption of frequency structures caused by data augmentation. Furthermore, by preserving only directional information, it maintains stable gradients even in the presence of significant discrepancies in frequency features. Diverging from traditional OT, which enforces global alignment based on Euclidean distance, SDE leverages entropy minimization on the hypersphere to achieve amplitude-robust gradient optimization and instance-level clustering that mitigates negative transfer.

B. Domain Adaptation for Time Series

Domain adaptation for time series has garnered significant attention due to its wide applications [15], [37], [38], [39].

For instance, VLH-DA aligns both local patterns and global sequence features via a virtual label hierarchy to address domain shifts [16]. A Granger causality alignment method is introduced to tackle conditional shifts in semi-supervised forecasting by uncovering cross-domain causal structures [40]. By generating augmented views through a temporal mixing strategy and leveraging contrastive learning, CoTMix [41] effectively reduces the distribution shift between the source and target domains, significantly improving the performance of unsupervised domain adaptation on time-series data. CLUDA [42] proposes the first contrastive learning framework that captures and aligns domain-invariant contextual representations in multivariate time series, significantly improving unsupervised domain adaptation performance. CALDA addresses multi-source UDA by combining contrastive and adversarial learning to leverage cross-source label information [17]. DACAD extends contrastive domain adaptation to anomaly detection by incorporating anomaly injection and dual contrastive objectives [43]. SEA++ [44] is developed as a multigraph-based higher-order sensor alignment framework in which local and global sensor-level features and correlations are jointly aligned to improve unsupervised domain adaptation for multivariate time-series data. B2TSDA introduces a domain adaption framework utilizing distillation and Shapley-enhanced weighting to address privacy and low-SNR challenges [45].

Nevertheless, these methods primarily focus on closed-set scenarios and struggle with private classes in UOSDA. TSFA

addresses this limitation by establishing an OSAM between the target samples and source prototypes, and adaptively adjusting the private-decision boundary.

III. METHOD

A. Problem Definition and Overview

In this article, the UOSDA for time-series classification is investigated. The assumption is that there exists a source-domain dataset $\mathcal{D}^s = (\mathbf{x}_i^s, y_i^s)_{i=1}^{n_s}$ and a target-domain dataset $\mathcal{D}^t = (\mathbf{x}_j^t)_{j=1}^{n_t}$, where $\mathbf{x}_i^s, \mathbf{x}_j^t \in \mathcal{X}$, $\mathcal{X} \subset \mathbb{R}^{n \times p}$ represent the time series of n time steps with p dimensions, and n_s and n_t are the number of samples in source and target domains, respectively. The label of source-domain data is $y_i^s, y_i^s \in \mathcal{Y}^s = \{1, \dots, C^s\}$. $\mathcal{Y}^s$ and $\mathcal{Y}^t$ denote the label sets of the source and target domains, respectively. The common label set is $\mathcal{Y}^s = \mathcal{Y}^s \cap \mathcal{Y}^t$. The data distributions of the two datasets are different, i.e., $P^s(\mathcal{X}) \neq P^t(\mathcal{X})$, where $P^s(\mathcal{X})$ is the marginal distribution of the source domain and $P^t(\mathcal{X})$ of the target domain. The conditional distributions across domains may also change, $P^s(\mathcal{Y}|\mathcal{X}) \neq P^t(\mathcal{Y}|\mathcal{X})$.

The objective of UOSDA is to train a feature extractor $\mathcal{F}$ and a classifier $\mathcal{H}$ on the source-domain dataset $\mathcal{D}^s$ and transfer them to the target-domain dataset $\mathcal{D}^t$ with the minimal generalization error $\mathbb{E}_{(\mathbf{x}^t, y^t) \sim P^t(\mathcal{X}, \mathcal{Y})} \mathcal{L}(\mathcal{H} \circ \mathcal{F}(\mathbf{x}^t), y^t)$, such that:

- 1) $\mathcal{F}$ and $\mathcal{H}$ classify target samples into correct common classes in $\mathcal{Y}^s$ and
- 2) $\mathcal{F}$ and $\mathcal{H}$ recognize all target private samples as unknown.

Thus, in UOSDA, all private classes are treated as one class. The label set of the source domain is $\mathcal{Y}^s = \{y_c\}_{c=1}^{C^s}$ and the label set of the target domain is $\mathcal{Y}^t = \{y_c\}_{c=1}^{C^s+1}$, where the label y_{C^s+1} represents all the private target classes. The parameters of $\mathcal{F}$ and $\mathcal{H}$ are $\theta_{\mathcal{F}}$ and $\theta_{\mathcal{H}}$, respectively.

As illustrated in Fig. 3, the proposed TSFA framework is designed as a tightly coupled system to address the specific challenges of time-series UOSDA. It consists of four integrated components: 1) a time-frequency extractor to handle nonstationarity; 2) a standard cross-entropy loss $\mathcal{L}_{CE}$ to learn the discriminate representation of common classes in source domain; 3) a global alignment loss $\mathcal{L}_{GA}$ to align distributions at a coarse level by minimizing SDE; and 4) a local alignment loss $\mathcal{L}_{LA}$ to align features at a classwise level. Supporting these stages, a memory bank maintains historical representations to ensure the stability of global alignment. Meanwhile, an OSAM module leverages OT to generate accurate pseudolabels for $\mathcal{L}_{LA}$.

B. Feature Extractor

The feature extractor consists of two branches: one for frequency features and another based on convolutional neural networks (CNNs). Given an input sequence $x \in \mathbb{R}^{C \times L}$, the frequency features are extracted using the fast Fourier transform (FFT), resulting in the magnitudes of the top- K modes $\hat{z}^f \in \mathbb{R}^{C \times K}$. To extract domain-invariant components from the frequency features, we introduce a learnable weight

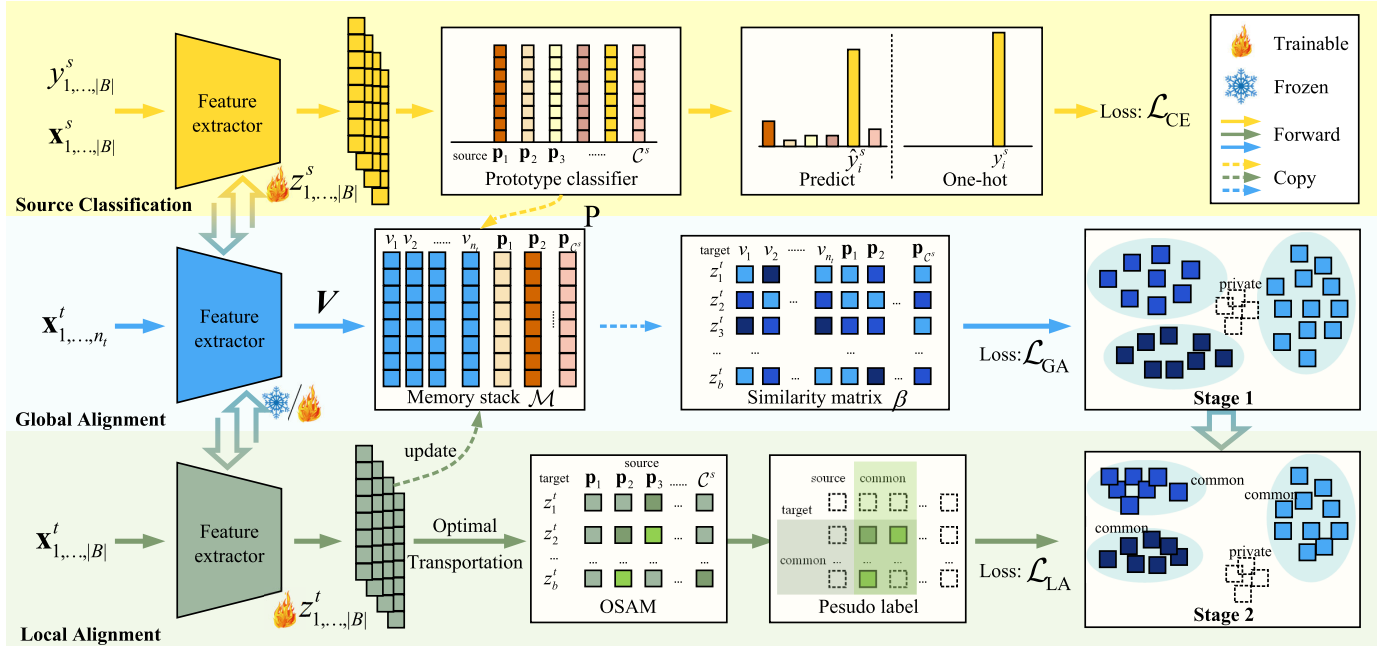


Fig. 3. Overview of TSFA. It contains source classification for minimizing the source risk, global alignment for shrinking the distribution discrepancy between source and target domains, and local alignment for mitigating the negative transfer and learning the domain-invariant representations of common classes. The shades of color in “Similarity matrix,” “OSAM,” and “Pseudolabel” have no special significance and are only meant to reflect value differences within the matrix.

$\mathbf{W} \in \mathbb{R}^{C \times C \times K}$, which is applied to the $\hat{\mathbf{z}}^f$ using an einsum operation

$$\mathbf{z}^f = \text{Avg}(\mathbf{W} \odot_K \hat{\mathbf{z}}^f)_C \quad (1)$$

where $\odot_K$ represents an elementwise weighted summation along the K -dimension. $\text{Avg}(\cdot)_C$ denotes averaging along the C -dimension. During the training process, $\mathbf{W}$ is optimized via backpropagation, where gradients derived from stable frequency modes across domains accumulate constructively, whereas gradients from domain-specific variations tend to cancel out. Consequently, this mechanism enables $\mathbf{W}$ to adaptively emphasize domain-invariant spectral components while suppressing task-irrelevant noise.

For the convolutional branch, each CNN block consists of a convolution layer, batch normalization (BN), ReLU activation, max pooling, and dropout. The forward propagation of each layer can be expressed as

$$\mathbf{z}^l = \text{ReLU}(\text{BN}(\mathbf{W}^l * \mathbf{z}^{l-1})) \in \mathbb{R}^{d_l \times l'} \quad (2)$$

where $\mathbf{W}^l$ represents the filter at layer l , $\mathbf{z}^{l-1}$ is the input from the previous layer, and d_l denotes the output dimension at layer l .

To obtain the temporal features, global average pooling (GAP) is applied to the feature map of the final convolutional layer along the time dimension

$$\mathbf{z}^t = \text{GAP}(\mathbf{z}^{l_{\text{last}}}) \in \mathbb{R}^{d'} \quad (3)$$

where $\mathbf{z}^{l_{\text{last}}}$ is the feature map from the last layer l_{last} and d' represents the dimensionality of the temporal features after pooling.

Finally, the time–frequency features are obtained by concatenating the two branches, resulting in $\mathbf{z} = [\mathbf{z}^t, \mathbf{z}^f]$.

In addition, a prototype-based classifier $\mathcal{H}$ is used in this work. Specifically, the classifier $\mathcal{H}$ consists of the

L2-normalized weights $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_{C^s}] \in \mathbb{R}^{d \times C^s}$, where d is the dimension of feature generated by $\mathcal{F}$, and C^s is the number of source classes. For the input feature $\mathbf{z}$, the probability that $\mathbf{z}$ belongs to the i th class is $\mathbf{q}_i = \text{softmax}(\hat{\mathbf{z}} \cdot \mathbf{p}_i)$, where $\hat{\mathbf{z}}$ is the L2-normalized $\mathbf{z}$.

C. Global Alignment

1) *Source-Domain Classification:* The classification error is minimized to train $\mathcal{F}$ and $\mathcal{H}$, enabling $\mathcal{H}$ to more accurately represent each source prototype. In each mini-batch $\mathbf{B}$, the output of $\mathcal{F}$ is fed into $\mathcal{H}$, and the predicted probabilities $\mathbf{q}_i$ are obtained through a softmax function. Then, the parameters $\theta_{\mathcal{F}}$ and $\theta_{\mathcal{H}}$ of $\mathcal{F}$ and $\mathcal{H}$ are updated using the multiclass cross-entropy loss $\mathcal{L}_{CE}$ between the one-hot vector $\mathbf{y}_i^s = \mathbf{1}(j = y_i^s)_{j=1}^{C^s}$ of class y_i^s and probabilities $\mathbf{q}_i$

$$\mathcal{L}_{CE}(\mathbf{y}_i^s, \mathbf{q}_i) = - \sum_{i=1}^{|\mathbf{B}|} \sum_{j=1}^{C^s} \mathbf{y}_{i,j}^s \log \mathbf{q}_{i,j} \quad (4)$$

where $\mathbf{q}_{i,j}$ denotes the probability that sample i is classified as j th class.

2) *Global Alignment in Target Domain:* To achieve consistent global alignment without high computational costs, we construct a memory bank $\mathcal{M}$ that stores all target features $\mathbf{V} \in \mathbb{R}^{n_t \times d}$ and source prototypes $\mathbf{P} \in \mathbb{R}^{C^s \times d}$. Initially, $\mathcal{F}$ is pretrained on the source domain, and $\mathbf{V}$ is extracted with frozen weights. During training, $\mathcal{M}$ is updated batchwise to maintain current feature representations

$$\mathcal{M} = \text{Concat}(\mathbf{V}, \mathbf{P}) \quad (5)$$

where all vectors are ℓ_2 -normalized to project features onto a unit hypersphere. This normalization ensures that the alignment relies solely on directional similarity, thereby mitigating

gradient instability caused by amplitude discrepancies in frequency features.

For a mini-batch of target features $\mathbf{B}$, we calculate the SDE over the memory bank. To prevent trivial solutions (self-matching), the current batch samples are masked out from $\hat{\mathcal{M}}$, denoted as $\hat{\mathcal{M}}$. The probability distribution $\beta_{i,j}$ for sample $\mathbf{z}_i^t$ matching the j th entry in $\hat{\mathcal{M}}$ is defined as

$$\beta_{i,j} = \frac{\exp(\hat{\mathcal{M}}_j^\top \mathbf{z}_i^t / \tau)}{\sum_k \exp(\hat{\mathcal{M}}_k^\top \mathbf{z}_i^t / \tau)} \quad (6)$$

where τ is a temperature parameter controlling the concentration of the distribution. Specifically, a smaller τ sharpens the distribution to emphasize the most similar prototypes, while a larger τ yields a smoother probability field.

To mitigate negative transfer, the global alignment loss $\mathcal{L}_{GA}$ is proposed by minimizing the entropy of β . This objective encourages each target sample to confidently align with its most similar neighbor (either a source prototype or another target sample) rather than maintaining a uniform distribution

$$\mathcal{L}_{GA} = -\frac{1}{|\mathbf{B}|} \sum_{i \in \mathbf{B}} \sum_j \beta_{i,j} \log(\beta_{i,j}). \quad (7)$$

To theoretically elucidate the intrinsic mechanism driving this clustering effect, the gradient dynamics of the entropy objective with respect to the feature embedding $\mathbf{z}_i^t$ are analyzed. By applying the chain rule, the gradient is derived as follows:

$$\nabla_{\mathbf{z}_i^t} \mathcal{L}_{GA} \propto -\frac{1}{\tau} \sum_j \beta_{i,j} (\log \beta_{i,j} + H(\beta_i)) \cdot \hat{\mathcal{M}}_j. \quad (8)$$

This derivation reveals an intrinsic selective alignment mechanism governed by the weighting coefficient $(\log \beta_{i,j} + H(\beta_i))$. For common class samples, the high similarity with a specific prototype results in a positive coefficient, thereby generating an attractive gradient that minimizes the distance to the corresponding source prototype. Conversely, for private samples characterized by a flat similarity distribution (where no single prototype dominates), these coefficients become negative. This induces a repulsive effect against known prototypes, driving private samples to diverge from the source support and coalesce into distinct clusters.

Remark 1: Due to the introduction of frequency features, using traditional discrepancy losses, such as KL divergence or MMD, can lead to gradient vanishing or explosion, which affects training stability. Private samples may be erroneously aligned toward the known classes due to this alignment constraint, resulting in negative transfer. In contrast, by normalizing features onto the unit hypersphere, the SDE is driven solely by directional consistency, effectively decoupling it from amplitude variations. The subsequent entropy minimization induces a self-reinforcing clustering effect, sharpening the decision boundaries between common and private classes without requiring explicit negative sampling.

D. Local Alignment

Following global alignment, the feature extractor $\mathcal{F}$ has learned certain discriminative representations for target-domain samples. However, the domain-invariant representation

learning for common samples still requires further improvement. To this end, local alignment leverages the pseudolabels provided by OSAM to perform self-supervised learning on target common samples, thereby enhancing the classification accuracy of common classes.

1) *Optimal Similarity Assignment Matrix:* To mitigate the noise inherent in pseudolabels derived solely from local pairwise similarity (e.g., nearest neighbors), the OSAM is proposed. Unlike greedy approaches that assign labels independently for each sample, OSAM formulates pseudolabeling as a global OT problem. By enforcing global distributional constraints, it achieves a globally consistent alignment that effectively filters out outliers and ambiguous samples. The formal definition is presented in Theorem 1.

Theorem 1: Let $\mathbf{M}^{cosine}$ denote the cosine similarity matrix between target features $\mathbf{Z}^t$ and source prototypes $\mathbf{P}$. γ^* is defined as the solution to the following entropy-regularized maximization problem:

$$\gamma^* = \arg \max_{\gamma \in \Pi(\mu, \nu)} \sum_{i,j} \gamma_{ij} \mathbf{M}_{ij}^{cosine} + \eta H(\gamma) \quad (9)$$

where $\Pi(\mu, \nu)$ is the set of valid coupling matrices satisfying marginal constraints μ and ν , $H(\gamma)$ is the entropic regularization term, and η is the regularization coefficient. A higher value of γ_{ij}^* indicates a higher confidence that target sample i belongs to the class represented by prototype j .

Proof: Standard OT aims to minimize the total transport cost. We define the transport cost matrix $\mathbf{C}$ based on the cosine distance

$$\mathbf{C}_{ij}^{cosine} = 1 - \mathbf{M}_{ij}^{cosine}. \quad (10)$$

Substituting this into the standard OT formulation, we obtain

$$\begin{aligned} \gamma^* &= \arg \min_{\gamma \in \Pi(\mu, \nu)} \sum_{i,j} \gamma_{ij} \mathbf{C}_{ij} \\ &= \arg \min_{\gamma \in \Pi(\mu, \nu)} \sum_{i,j} \gamma_{ij} (1 - \mathbf{M}_{ij}^{cosine}) \\ &= \arg \min_{\gamma \in \Pi(\mu, \nu)} \underbrace{\sum_{i,j} \gamma_{ij}}_{\text{const}} - \sum_{i,j} \gamma_{ij} \mathbf{M}_{ij}^{cosine} \\ &= \arg \max_{\gamma \in \Pi(\mu, \nu)} \sum_{i,j} \gamma_{ij} \mathbf{M}_{ij}^{cosine}. \end{aligned} \quad (11)$$

Since $\sum_{i,j} \gamma_{ij}$ is constant, minimizing the cosine cost is equivalent to maximizing the total similarity. Finally, to ensure strictly convex optimization and facilitate efficient computation via the Sinkhorn algorithm [46], the entropic regularization term $\eta H(\gamma)$ is incorporated, yielding the objective function defined in (9). ■

Remark 2: Different from previous works that utilize OT primarily as a discrepancy metric to align distributions, OSAM exploits the OT plan itself to generate pseudolabels. While standard cosine similarity captures only local pairwise relationships, OSAM accounts for the global distribution structure via the marginal constraints. This global optimization effectively prevents trivial solutions and enhances the distinction between common and private classes.

2) *Self-Supervised Local Alignment*: To obtain pseudolabels of the common samples in the target domain, the OSAM between the current mini-batch of target samples and source prototypes is established. First, the cost matrix $\mathbf{M}^{cosine}$ is obtained based on the cosine similarity between the mini-batch features of target samples $\mathbf{Z}^t = [\mathbf{z}_1^t, \mathbf{z}_2^t, \dots, \mathbf{z}_{|B|}^t]$ and the source prototypes $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_{C^s}]$. Then, the optimal transportation distances for the elements in $\mathbf{P}$ and $\mathbf{Z}^t$ are calculated. To facilitate the description, the calculation of the OSAM in (9) is simplified as follows:

$$\gamma^* \leftarrow \text{OSAM}(\mathbf{M}^{cosine}, \mu, \nu) \quad (12)$$

where $\mathbf{M}^{cosine} = \mathbf{Z}^t \mathbf{P}^T$ represents the cosine similarity between the target-instance feature $\mathbf{Z}^t$ and source prototypes $\mathbf{P}$. The vector $\mu = (1/|B|)\mathbf{1}$ and $\nu = (1/C^s)\mathbf{1}$.

Intuitively, within matrix γ^* , private classes are likely to be assigned relatively lower weights and exhibit low similarity to common class prototypes, thus increasing the likelihood of being identified as private classes. To avoid negative transfer to private samples, local alignment should only be applied to common class samples with higher weights.

Then, γ^* is normalized to obtain $\tilde{\gamma}^*$, where the i th row in $\tilde{\gamma}^*$ can be viewed as the predicted probability for the i th instance in the mini-batch.

The pseudolabel $\hat{y}_i^t$ of target samples i mini-batch can be obtained

$$\hat{y}_i^t = \arg \max (\{\tilde{\gamma}_{i,1}^*, \tilde{\gamma}_{i,2}^*, \dots, \tilde{\gamma}_{i,C^s}^*\}). \quad (13)$$

However, in certain extreme cases, private samples in the target domain may be incorrectly classified as common classes. To address this issue, two statistical metrics are calculated

$$\omega_j^s = \sum_{i=1}^B \tilde{\gamma}_{i,j}^*, \quad \omega_i^t = \sum_{j=1}^{C^s} \tilde{\gamma}_{i,j}^* \quad (14)$$

where ω_j^s and ω_i^t represent the importance of the source prototype $\mathbf{p}_j$ and the target instance $\mathbf{z}_i^t$, respectively. A higher ω_j^s indicates that $\mathbf{p}_j$ is frequently assigned to target samples, suggesting that $\mathbf{p}_j$ belongs to a common class in both the target and source domains. Similarly, a higher ω_i^t signifies that $\mathbf{z}_i^t$ is frequently assigned to source prototypes, indicating that $\mathbf{z}_i^t$ belongs to the common classes. Based on these insights, adaptive decision boundaries $(1/b)$ and $(1/C^s)$ are utilized. Using these decision boundaries, a Kronecker delta function δ is obtained

$$\delta_i = \begin{cases} 1, & \omega_i^t \geq \frac{1}{|B|} \text{ and } \omega_{\hat{y}_i^t}^s \geq \frac{1}{C^s} \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

where $\delta_i = 1$ indicates that the instance $\mathbf{x}_i^t$ is identified as a common class with high confidence and can be assigned to a pseudolabel $\hat{y}_i^t$ by (13). The cross-entropy loss for the common classes is then calculated using these pseudolabels, facilitating their local alignment

$$\mathcal{L}_{LA} = \frac{\sum_{i=1}^{|B|} \delta_i \cdot \mathcal{L}_{CE}(\tilde{\gamma}_i^*, \hat{y}_i^t)}{\sum_{i=1}^{|B|} \delta_i}. \quad (16)$$

By minimizing the local alignment loss $\mathcal{L}_{LA}$, the features of common and private samples are significantly distinguished,

Algorithm 1 Complete Steps of TSFA

Require: Source dataset $\mathcal{D}^s = (\mathbf{x}_i^s, y_i^s)_{i=1}^{n_s}$, target dataset $\mathcal{D}^t = (\mathbf{x}_j^t)_{j=1}^{n_t}$

- 1: **Initialize:** Randomly initialize $\theta_{\mathcal{H}}$ with C^s output dimensions
- 2: **step1: pretrain**
- 3: Initialize memory bank $\mathcal{M}$ with $\mathcal{F}(\mathcal{D}^t)$.
- 4: Initialize classifier weights $\theta_{\mathcal{H}}$ (Prototypes $\mathbf{P}$).
- 5: **step2: train**
- 6: **for** Epoch $e = 1$ to E **do**
- 7: **for** Minibatch $b = 1$ to B **do**
- 8: $\mathbf{x}^s, y^s \leftarrow \mathcal{D}^s$, $\mathbf{x}^t, id^t \leftarrow \mathcal{D}^t$;
- 9: Forward: $\mathbf{z}^s, \mathbf{z}^t \leftarrow \mathcal{F}(\mathbf{x}^s), \mathcal{F}(\mathbf{x}^t)$;
- 10: Classification loss: $\mathcal{L}_{CE}$ according to (4);
- 11: Prototypes: $\mathbf{P} \leftarrow$ the L2-normalized weights of $\mathcal{H}$;
- 12: Similarity matrix: β according to (6);
- 13: Global loss: $\mathcal{L}_{GA}$ according to (7);
- 14: **if** $b < \frac{B}{2}$ **then**
- 15: Update $\theta_{\mathcal{F}}, \theta_{\mathcal{H}}$ to $\arg \min \mathcal{L}_{CE} + \lambda_1 \mathcal{L}_{GA}$;
- 16: **end if**
- 17: **if** $b \geq \frac{B}{2}$ **then**
- 18: OSAM: $\gamma^* \leftarrow \text{OSAM}(\mathbf{M}^{cosine}, \frac{1}{B}\mathbf{1}, \frac{1}{C^s}\mathbf{1})$;
- 19: Pseudo label: $\hat{y}^t \leftarrow \arg \max(\gamma^*, \dim = 1)$;
- 20: Delta function: $\delta \leftarrow \omega_i^t \geq \frac{1}{B}$ and $\omega_{\hat{y}_i^t}^s \geq \frac{1}{C^s}$;
- 21: Local loss: $\mathcal{L}_{LA}$ according to (16);
- 22: $\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda_1 \mathcal{L}_{GA} + \lambda_2 \mathcal{L}_{LA}$;
- 23: Update $\theta_{\mathcal{F}}, \theta_{\mathcal{H}}$ to $\arg \min \mathcal{L}_{total}$;
- 24: **end if**
- 25: Update $\mathcal{M}$ with $\mathbf{P}$ and $\mathbf{z}^t$ corresponding to id^t ;
- 26: **end for**
- 27: **end for**

and the classifier can accurately identify private classes using adaptive thresholds.

3) *Private Classes Recognition*: Inspired by the above analysis, a private-class recognition approach based on OSAM is proposed. The idea of new-class recognition is already incorporated into the self-supervised alignment, and a clear statement of it is provided in this section.

First, the OSAM γ between the target samples and source prototypes is built upon (12), and it is then L2-normalized to obtain γ^* . Next, based on the dimensions of the OSAM, the similarity threshold for new-class recognition is set to $1/C^s$. Finally, private samples are identified based on the threshold $1/C^s$

$$y_i^t = \begin{cases} \arg \max (\tilde{\gamma}_{i,1}^*, \tilde{\gamma}_{i,2}^*, \dots, \tilde{\gamma}_{i,C^s}^*), & q_i^t > \frac{1}{C^s} \\ C^s + 1, & \text{otherwise} \end{cases} \quad (17)$$

where $q_i^t = \max(\tilde{\gamma}_{i,1}^*, \tilde{\gamma}_{i,2}^*, \dots, \tilde{\gamma}_{i,C^s}^*)$, and all private classes are considered as C^s . Compared to traditional methods based on fixed confidence thresholds or additional classification heads, the proposed method is more flexible and robust.

E. Training and Inference

The complete pipeline of TSFA, summarized in Algorithm 1, follows a coarse-to-fine optimization strategy

TABLE I
DETAILED INFORMATION OF **HAR/CWRU/BEAR** DATASETS

Dataset	Domains	Classes	Channels	Length
HAR [47]	30	6	9	128
CWRU [48]	4	10	2	256
BEAR [49]	2	5	8	256

TABLE II
DOMAIN SEGMENTATION ON THE **HAR** DATASET

domain number	classes
1,2,3,4	walking, walking upstairs, walking downstairs, sitting, standing, laying
5,6,7,8	walking, walking upstairs, walking downstairs, sitting
9,10,11,12	walking, walking upstairs, walking downstairs
13,14,15,16	walking upstairs, walking downstairs, sitting, standing
17,18,19,20	walking downstairs, sitting, standing, laying
21,22	walking, walking upstairs, walking downstairs, standing
23,24,25,26	walking, walking downstairs, sitting, standing
27,28	walking downstairs, sitting, standing
29,30	walking upstairs, standing, laying

designed to ensure stability. To handle nonstationarity, the feature extractor $\mathcal{F}$ is pretrained on the source domain. A memory bank $\mathcal{M}$ is then initialized with target features to support consistent global alignment. In the initial phase (for mini-batch index $b \leq B/2$), we optimize only the source classification loss $\mathcal{L}_{CE}$ and the global alignment loss $\mathcal{L}_{GA}$. This warm-up period allows the source prototypes $\mathbf{P}$ (weights of $\mathcal{H}$) to form meaningful cluster centers before they are used for pseudolabeling. Once the prototypes stabilize ($b > B/2$), the OSAM module is activated. It calculates the OT plan between target features $\mathbf{z}^t$ and prototypes $\mathbf{P}$ to generate high-quality pseudolabels, enabling the local alignment loss $\mathcal{L}_{LA}$. The total objective function is formulated as

$$\arg \min_{\theta_{\mathcal{F}}, \theta_{\mathcal{H}}} \sum_{\mathbf{x}^s, \mathbf{y}^s \in \mathcal{D}^s, \mathbf{x}^t \in \mathcal{D}^t} \mathcal{L}_{CE}(\mathcal{H} \circ \mathcal{F}(\mathbf{x}^s), \mathbf{y}^s) \quad (18)$$

$$+ \lambda_1 \mathcal{L}_{GA}(\mathcal{F}(\mathbf{x}^t), \mathbf{P}) \\ + \lambda_2 \mathcal{L}_{LA}(\mathcal{F}(\mathbf{x}^t), \mathbf{P}) \quad (19)$$

where λ_1 and λ_2 are hyperparameters that weight the importance of global and local alignment.

During testing, the process comprises three steps.

- 1) *Feature Extraction*: Compute features $\mathbf{z}^t = \mathcal{F}(\mathbf{x}^t)$.
- 2) *Alignment*: Compute the OSAM transport plan between $\mathbf{z}^t$ and $\mathbf{P}$.
- 3) *Decision*: Perform open-set classification based on (17) using the threshold $1/C^s$.

IV. EXPERIMENTS

A. Datasets and Settings

TSFA is evaluated on three real-world cross-domain time-series datasets: **HAR** [47], **CWRU** [48], and **BEAR** [49]. **HAR**: This dataset contains motion data (acceleration and

TABLE III
DOMAIN SEGMENTATION ON THE **CWRU** DATASET

domain number	operating conditions	classes
0	0HP	B007, B014, B021, IR007, IR014, IR021, OR007, OR014, OR021, normal
1	1HP	B007 B014 B021, IR007, IR014, IR021, OR007, OR014
2	2HP	B007 B014 B021, IR007, IR014, IR021
3	3HP	B007 B014 B021, IR007, IR014

TABLE IV
DOMAIN SEGMENTATION ON THE **BEAR** DATASET

domain number	operating conditions	classes
0	20Hz-0V	ball, comb, health, inner, outer
1	30Hz-20V	ball, comb, health, inner, outer
2	20Hz-0V	ball, comb, health
3	30Hz-20V	ball, comb, health, inner
4	20Hz-0V	health, inner, outer
5	30Hz-20V	health, inner, outer
6	20Hz-0V	health, comb, inner, outer
7	30Hz-20V	ball, comb, inner, outer

angular velocity) from 30 volunteers performing six activities (e.g., walking and laying). Since biometric motion patterns vary significantly by individual, each subject is treated as an independent domain, resulting in 30 domains. **CWRU**: Collected from motor bearings under varying loads (0–3 HP) and speeds (1720–1797 r/min). We define domains based on the *motor load*, as load changes fundamentally alter vibration amplitudes and resonant frequencies. Data was downsampled to 12 kHz to unify sampling rates. **BEAR**: Simulates fan gear faults under different speed and load configurations. Domains are defined by the *operating condition pairs*: 20 Hz/0 V and 30 Hz/2 V. In the preprocessing stage, the raw data are segmented using a sliding-window approach, followed by min–max normalization to scale the features into the range of [0, 1]. The detailed information of the HAR, CWRU, and BEAR datasets is summarized in Table I.

Detailed segmentation settings and class allocations for open-set scenarios are provided in Tables II, III, and IV. The Kolmogorov–Smirnov (KS) test is employed to quantify the distributional discrepancy between domains. Under the domain-segmentation strategy adopted in this article, all scenarios consistently yield p values < 0.05 , demonstrating a significant interdomain divergence. Openness is calculated as $1 - |\mathcal{Y}_s \cap \mathcal{Y}_t|/|\mathcal{Y}_t|$. All experiments were conducted on an Intel Core i7-11700K CPU and NVIDIA RTX 3080 Ti GPU using PyTorch 2.1.1. To ensure fair comparison, all baseline methods were reimplemented using the identical feature extractor and preprocessing steps described above.

To evaluate the performance of the UOSDA algorithm, the following five metrics are employed: acc_{total} denotes the classification accuracy of all samples in the target domain, acc_{pri} represents the accuracy of correctly identifying private samples, and acc_{com} refers to the classification accuracy of common classes. In UOSDA algorithms, it is crucial to con-

TABLE V
 acc_{total} AND acc_{com} COMPARISON OF THE UOSDA TASKS ON HAR/CWRU/BEAR DATASETS

scenarios		OVANet [11]		KAUNiDA [28]		STUN [30]		RAINCOAT [18]		GLC++ [31]		COSDA [33]		TFSA	
		acc_{total}	acc_{com}	acc_{total}	acc_{com}	acc_{total}	acc_{com}	acc_{total}	acc_{com}	acc_{total}	acc_{com}	acc_{total}	acc_{com}	acc_{total}	acc_{com}
HAR	5 to 4	31.25	45.25	59.38	86.51	62.50	87.50	67.18	82.92	67.19	79.02	54.69	85.63	86.98	83.72
	6 to 3	32.19	25.00	64.06	75.02	54.69	64.48	68.12	58.70	53.62	75.00	57.97	75.00	87.09	88.21
	9 to 8	37.50	25.93	37.50	19.05	65.50	89.68	42.86	43.47	40.00	60.32	34.29	47.09	96.43	95.24
	10 to 7	56.25	38.69	43.75	64.00	69.63	78.70	63.41	60.61	56.10	60.80	46.34	61.11	83.59	77.88
	10 to 21	21.88	33.80	40.63	59.09	76.37	85.18	51.06	65.52	40.43	66.67	44.68	73.33	95.83	94.44
	11 to 22	34.38	66.67	53.13	66.67	59.37	100.00	57.89	68.18	52.63	91.67	65.79	86.31	93.75	91.67
	average	35.58	39.22	49.74	61.72	64.68	84.26	58.42	63.23	51.66	72.25	50.63	71.41	90.61	88.53
CWRU	1 to 0	38.95	35.56	46.09	27.69	42.96	41.50	44.64	52.32	85.86	97.98	82.87	99.30	77.98	75.23
	2 to 0	28.68	31.56	71.43	68.65	28.12	39.81	46.63	71.28	50.61	56.90	55.80	78.18	92.67	91.23
	2 to 1	36.82	32.11	67.34	65.51	45.46	61.51	62.59	82.92	75.59	99.81	78.08	98.47	78.04	74.47
	3 to 0	14.06	12.50	62.95	38.00	13.05	20.00	37.90	27.84	34.58	12.83	50.50	73.66	86.96	84.35
	3 to 1	15.22	20.00	58.44	38.21	16.41	26.35	47.38	65.62	47.63	30.28	61.15	73.73	84.28	81.29
	3 to 2	18.57	0.00	64.65	56.53	53.12	66.56	38.52	43.63	81.96	100.00	88.97	98.11	80.07	76.09
	average	25.38	21.96	61.82	49.10	33.19	42.62	46.28	57.27	62.70	66.30	69.56	86.91	83.33	80.44
BEAR	2 to 1	52.29	87.29	25.62	39.23	21.17	0.63	42.48	28.41	51.20	86.12	77.92	93.08	97.09	96.11
	2 to 3	34.50	15.09	24.44	33.33	28.12	39.81	41.82	23.39	85.45	95.02	69.09	62.60	91.48	88.64
	3 to 0	30.47	21.81	65.80	72.16	43.04	50.00	48.53	54.75	78.68	93.53	77.33	96.58	79.94	74.91
	5 to 6	51.00	33.17	44.81	57.14	25.78	33.33	55.61	72.71	50.58	66.50	50.43	65.13	82.06	76.45
	6 to 1	67.72	65.77	41.58	42.62	45.18	26.10	46.24	52.60	80.12	96.45	82.84	97.33	89.18	86.48
	7 to 0	21.53	24.77	36.11	43.56	33.14	41.23	69.58	68.10	54.17	65.27	60.12	74.89	59.52	50.57
	average	42.92	41.32	39.73	48.01	32.74	31.85	50.71	49.99	66.70	83.82	69.62	81.60	83.21	78.86

TABLE VI
 acc_{pri} AND H-SCORE COMPARISON OF THE UOSDA TASKS ON HAR/CWRU/BEAR DATASETS

scenarios		OVANet [11]		KAUNiDA [28]		STUN [30]		RAINCOAT [18]		GLC++ [31]		COSDA [33]		TFSA	
		acc_{pri}	H-score	acc_{pri}	H-score	acc_{pri}	H-score	acc_{pri}	H-score	acc_{pri}	H-score	acc_{pri}	H-score	acc_{pri}	H-score
HAR	5 to 4	7.14	12.15	8.70	15.81	21.42	34.43	39.13	53.17	78.26	78.64	21.74	34.68	95.65	91.14
	6 to 3	100.00	40.00	54.54	63.16	80.00	71.41	86.96	70.08	30.43	43.29	47.82	58.40	82.61	85.32
	9 to 8	50.00	34.15	72.73	30.19	27.84	42.49	41.67	42.55	25.00	35.35	33.33	39.03	100.00	97.56
	10 to 7	100.00	29.81	0.00	0.00	14.44	24.40	75.00	67.04	0.00	0.00	0.00	0.00	100.00	87.71
	10 to 21	0.00	0.00	0.00	0.00	60.67	70.87	27.78	39.01	0.00	0.00	0.00	0.00	100.00	97.14
	11 to 22	0.00	0.00	20.00	30.77	0.00	0.00	43.75	53.30	0.00	0.00	50.00	63.32	100.00	95.65
	average	42.86	19.35	26.00	23.32	34.06	40.60	52.38	54.19	22.28	26.21	25.48	32.57	96.38	92.42
CWRU	1 to 0	96.93	52.03	4.73	8.08	75.75	53.62	6.00	10.77	78.67	87.27	51.33	67.68	100.00	85.86
	2 to 0	96.29	47.98	71.88	70.23	12.50	19.02	3.06	5.86	67.76	61.86	34.03	47.42	98.45	94.71
	2 to 1	100.00	48.61	78.05	71.23	2.53	4.86	3.09	5.96	17.74	30.13	51.08	67.27	100.00	83.75
	3 to 0	26.53	16.99	93.57	54.05	27.17	23.04	49.52	32.63	69.11	21.64	51.03	60.29	100.00	91.03
	3 to 1	19.54	19.77	94.96	54.49	63.51	30.42	16.84	26.80	94.96	45.92	43.17	54.46	99.26	89.75
	3 to 2	100.00	0.00	96.91	71.41	61.05	30.13	16.67	24.12	15.15	26.31	100.00	99.05	100.00	86.56
	average	73.22	30.90	73.35	54.92	40.42	26.85	15.86	17.69	57.23	45.52	55.11	66.03	99.62	88.61
BEAR	2 to 1	1.90	3.72	5.13	9.07	100.00	1.26	63.43	39.24	0.06	0.12	69.62	79.66	100.00	97.98
	2 to 3	95.47	26.06	0.00	0.00	26.23	24.40	97.54	37.73	54.03	68.89	88.83	73.44	100.00	93.98
	3 to 0	72.30	33.51	41.17	52.43	36.17	41.98	23.52	63.91	18.60	31.03	0.00	0.00	100.00	85.67
	5 to 6	100.00	49.81	5.98	10.82	0.00	0.00	6.82	12.47	8.05	14.36	11.28	19.23	98.89	86.24
	6 to 1	71.08	68.33	34.12	37.90	100.00	41.40	18.04	26.86	11.80	21.03	22.21	36.17	100.00	92.75
	7 to 0	1.06	2.03	3.11	5.80	0.00	0.00	76.13	71.89	12.96	21.63	0.86	1.70	95.33	66.09
	average	56.97	30.58	14.92	19.34	43.73	18.17	47.58	42.02	17.58	26.18	32.13	35.03	99.04	87.12

sider the tradeoff between the accuracy of common classes and that of private classes. Therefore, H-Score is designed to represent the harmonic mean of acc_{com} and acc_{pri}

$$H - Score = \frac{2acc_{com}acc_{pri}}{acc_{com} + acc_{pri}}. \quad (20)$$

B. Comparison to State-of-the-Art Methods

TSFA is compared against six representative UOSDA methods: OVANet [11], KAUNiDA [28], STUN [30], RAINCOAT [18], GLC++ [31], and COSDA [33]. Results across three datasets are summarized in Tables V and VI.

TSFA consistently achieves the best overall performance, with average acc_{total} of 90.61%, 83.33%, and 83.21% on the HAR, CWRU, and BEAR datasets, respectively. On HAR, it maintains high accuracy despite the challenge of distinguishing mixed dynamic and static activities. On CWRU, TSFA exhibits superior stability across varying openness, particularly in complex multiclass scenarios (e.g., 2 to 0). On BEAR, while the “comb” fault class poses inherent difficulty, TSFA still significantly outperforms baselines.

Notably, TSFA consistently yields the highest H-score, demonstrating an effective balance between common- and private-class recognition. While baselines often suffer from

TABLE VII
ABLATION RESULTS acc_{TOTAL} ON HAR/CWRU/BEAR DATASETS

Datasets	Modules		Scenarios					
	$\mathcal{L}_{GA}$	$\mathcal{L}_{LA}$	5 to 4	6 to 3	9 to 8	10 to 7	10 to 21	11 to 22
HAR	$\times$	$\times$	46.49	54.78	80.80	53.58	73.57	51.90
	$\checkmark$	$\times$	69.28	62.67	48.81	58.03	63.54	77.08
	$\times$	$\checkmark$	52.78	65.75	72.77	50.41	48.63	52.50
	$\checkmark$	$\checkmark$	86.98	87.09	96.43	83.59	95.83	93.75
	$\mathcal{L}_{GA}$	$\mathcal{L}_{LA}$	1 to 0	2 to 0	2 to 1	3 to 0	3 to 1	3 to 2
CWRU	$\times$	$\times$	67.13	69.54	65.98	38.06	42.20	50.86
	$\checkmark$	$\times$	66.37	76.79	68.59	37.25	47.31	35.62
	$\times$	$\checkmark$	71.03	67.31	69.69	47.48	55.07	62.19
	$\checkmark$	$\checkmark$	77.98	92.67	78.04	86.96	84.28	80.07
	$\mathcal{L}_{GA}$	$\mathcal{L}_{LA}$	2 to 1	2 to 3	3 to 0	5 to 0	6 to 1	7 to 0
BEAR	$\times$	$\times$	76.76	66.30	73.68	68.52	43.01	40.15
	$\checkmark$	$\times$	96.43	91.05	80.78	65.55	80.23	32.75
	$\times$	$\checkmark$	96.22	90.93	80.64	47.67	69.96	29.88
	$\checkmark$	$\checkmark$	97.09	91.48	79.94	82.06	89.18	59.52

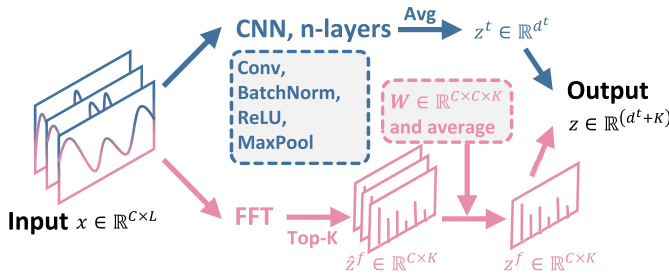


Fig. 4. Architecture of time-frequency feature extractor.

negative transfer or fail to identify private samples (sometimes yielding zero acc_{pri}). TSFA maintains acc_{pri} above 80% in most cases. This advantage is primarily attributed to the OSAM-based local alignment and the adaptive decision boundary, which together ensure robust rejection of private samples while preserving high discriminability for common classes.

C. Ablation Study

1) *Effectiveness of TSFA Framework*: As shown in Table VII, the contribution of global and local alignment is evaluated. Compared to the baseline [source only ($\times, \times$)], incorporating global alignment ($\checkmark, \times$) alone improves acc_{total} by over 5%, effectively reducing domain uncertainty through coarse semantic alignment. Local alignment ($\times, \checkmark$) further enhances performance (nearly 4% gain) by refining decision boundaries for complex classes (e.g., in BEAR). Notably, while single-stage alignment may occasionally suffer from misalignment under severe shifts, the full TSFA ($\checkmark, \checkmark$) achieves the highest H -scores across all datasets, demonstrating that the two stages complement each other to ensure stable open-set recognition.

2) *Time-Frequency Features*: Table VIII compares TSFA using different feature inputs. The significant performance drop in “TSFA w/o frequency” and “TSFA w/o time” configurations validates that frequency-domain information provides crucial domain-invariant cues. Integrating both domains allows the model to better capture the nonstationary characteristics of time-series data.

3) *SDE Mitigates the Negative Transfer*: To verify the superiority of $\mathcal{L}_{GA}$, we replace it with traditional discrepancy

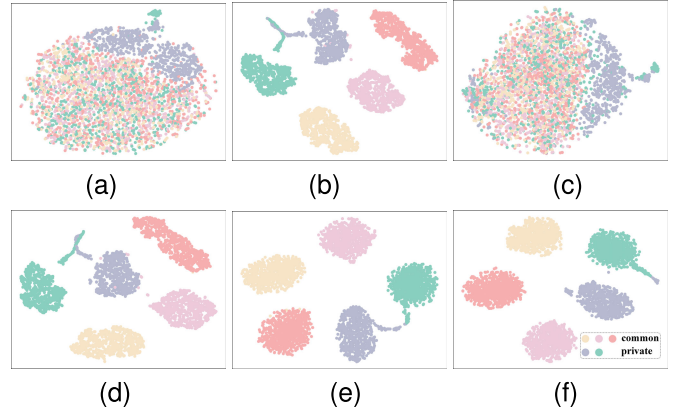


Fig. 5. t-SNE [50] visualization of features on “2-to-1” scenario in BEAR dataset. (a) No time. (b) No frequency. (c) $\mathcal{L}_{KL}$. (d) $\mathcal{L}_{MMD}$. (e) $\mathcal{L}_{OT}$. (f) $\mathcal{L}_{GA}$.

metrics (KL, MMD, and OT). As indicated in Table IX, these metrics—especially KL divergence—lead to performance degradation and negative transfer. This confirms that the cosine-based SDE is more robust for aligning time-frequency features than standard geometric or statistical distances.

4) *Features Visualization*: t-SNE [50] is utilized to qualitatively assess the feature distributions and identify potential negative transfer effects. As illustrated in Fig. 5, using only frequency-domain features or adopting KL divergence as the alignment criterion resulted in severe class overlap, indicating pronounced negative transfer. In contrast, when temporal features alone are employed, or when MMD or OT served as the alignment objective, only mild class confusion is observed, suggesting relatively weaker negative transfer. Notably, the proposed TSFA method yields the most discriminative feature representations, with clear class separation. These results substantiate the effectiveness of frequency-aware and globally aligned strategies specifically designed for temporal-domain adaptation tasks.

D. Analysis and Discussion

In this section, several experiments are conducted to analyze the factors influencing the performance of the TSFA methods, including the proportion of private classes among all categories, discriminative feature analysis, selection of hyperparameters, and visualization of feature alignment.

1) *Openness*: The robustness of TSFA against varying openness is evaluated on the CWRU dataset and compared with six SOTA methods (see Fig. 6). While the performance of baseline methods fluctuates or declines significantly as the number of private classes increases, TSFA consistently maintains high and stable acc_{total} . This demonstrates that the synergistic effect of our global and local alignment strategies effectively mitigates the negative impact of openness, ensuring robust performance across scenarios with varying degrees of difficulty.

2) *Discriminant Feature Analysis*: The 2-to-1, 3-to-1, and 3-to-2 scenarios in the CWRU dataset are selected to perform t-SNE [50] visualization. As shown in Fig. 7, the baseline model (source only) exhibits heavy cluster entanglement due to domain shifts, failing to distinguish common and private

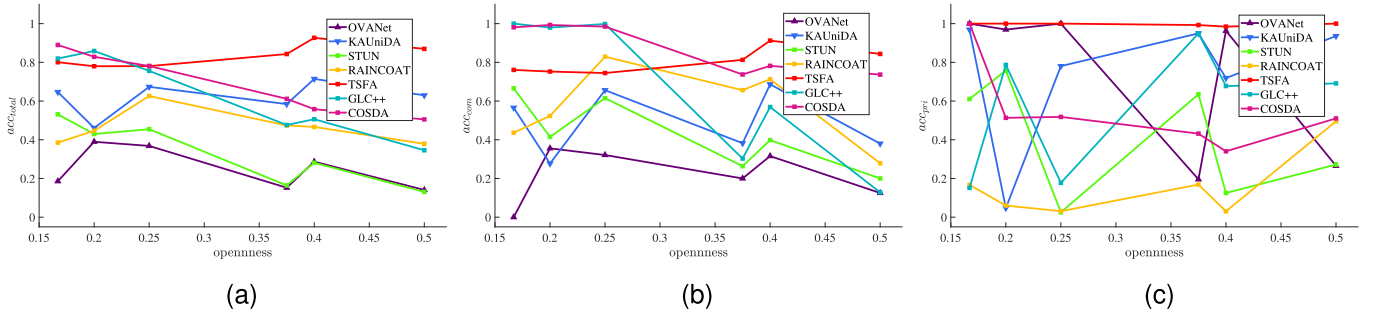


Fig. 6. acc_{total} , acc_{com} , and acc_{pri} varying openness on **CWRU** dataset. (a) Total accuracy. (b) Common accuracy. (c) Private accuracy.

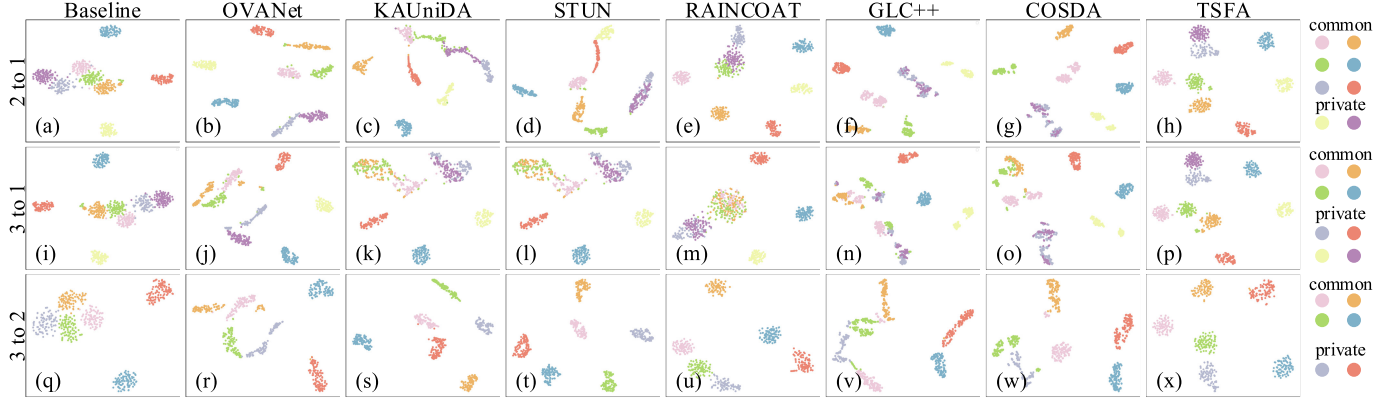


Fig. 7. t-SNE [50] visualization of the baseline, OVANet [11], KAUniDA [28], STUN [30], RAINCOAT [18], GLC++ [31], COSDA [33], and TSFA on **CWRU** dataset. (a)–(h) Visualization of scenarios “2 to 1.” (i)–(p) Visualization of scenarios “3 to 1.” (q)–(x) Visualization of scenarios “3 to 2.”

TABLE VIII

AVERAGE TOTAL ACCURACY AND H-SCORE W/O FREQUENCY FEATURES

features	HAR		CWRU		BEARING	
	acc_{total}	H-score	acc_{total}	H-score	acc_{total}	H-score
TSFA w/o frequency	43.23	38.52	45.40	52.30	50.11	52.26
TSFA w/o time	44.64	44.86	36.31	41.12	38.14	33.65
TSFA	90.61	92.42	83.33	88.61	83.21	87.12

TABLE IX

AVERAGE TOTAL ACCURACY AND H-SCORE OF TSFA WITH DIFFERENT FEATURE ALIGNING METRICS

loss	HAR		CWRU		BEARING	
	acc_{total}	H-score	acc_{total}	H-score	acc_{total}	H-score
$\mathcal{L}_{KL}$	35.52	24.51	79.72	86.05	31.54	21.79
$\mathcal{L}_{MMD}$	87.59	89.85	82.67	88.48	64.08	67.01
$\mathcal{L}_{OT}$	87.23	89.54	82.64	88.41	64.32	67.92
$\mathcal{L}_{GA}$	90.61	92.42	83.33	88.61	83.21	87.12

classes. While methods like OVANet [11] and GLC++ [31] enlarge interclass distances, they still suffer from confusion at the private–common boundaries. Metrics-based methods, such as KAUniDA [28] and RAINCOAT [18], show limited efficacy on time-series data, as they rely on large-scale samples for hyperparameter tuning or statistical tests (e.g., Dip test). Similarly, STUN [30] and COSDA [33] improve generalization through stochastic networks or prototypes but still struggle with precise separation in complex fault scenarios.

In contrast, TSFA consistently exhibits the most distinct class separation and stable boundaries. The visualization con-

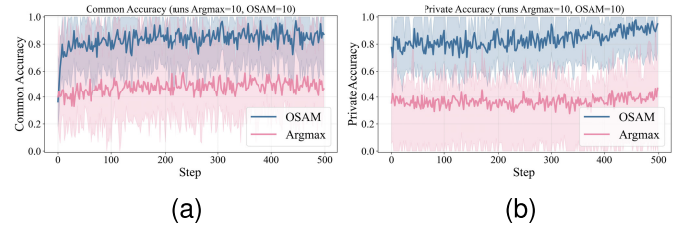


Fig. 8. Per-batch acc_{com} and acc_{pri} under OSAM and Argmax strategies on the CWRU 3-to-0 scenario during training. (a) acc_{com} . (b) acc_{pri} .

firms that: 1) high classification accuracy in Tables V and VI directly correlates with stronger feature discriminability and 2) the SDE-based global alignment and OSAM-based local alignment synergistically enhance the feature extractor’s ability to isolate private samples while maintaining tight clusters for common classes. This qualitative evidence substantiates TSFA’s superior robustness in cross-domain time-series classification.

3) *Analysis of Pseudolabel Reliability*: To verify reliability, acc_{com} and acc_{pri} are monitored throughout training, averaged over ten independent runs (see Fig. 8). Compared to the Argmax baseline (using the same $1/C^s$ threshold), OSAM demonstrates superior robustness. In early epochs, OSAM exhibits smoother trajectories and higher accuracy, indicating that global constraints effectively correct local misalignments despite limited feature quality. Subsequently, acc_{com} under OSAM stabilizes at $> 90\%$ while Argmax fluctuates, confirming that confidence screening filters unreliable samples to prevent error propagation.

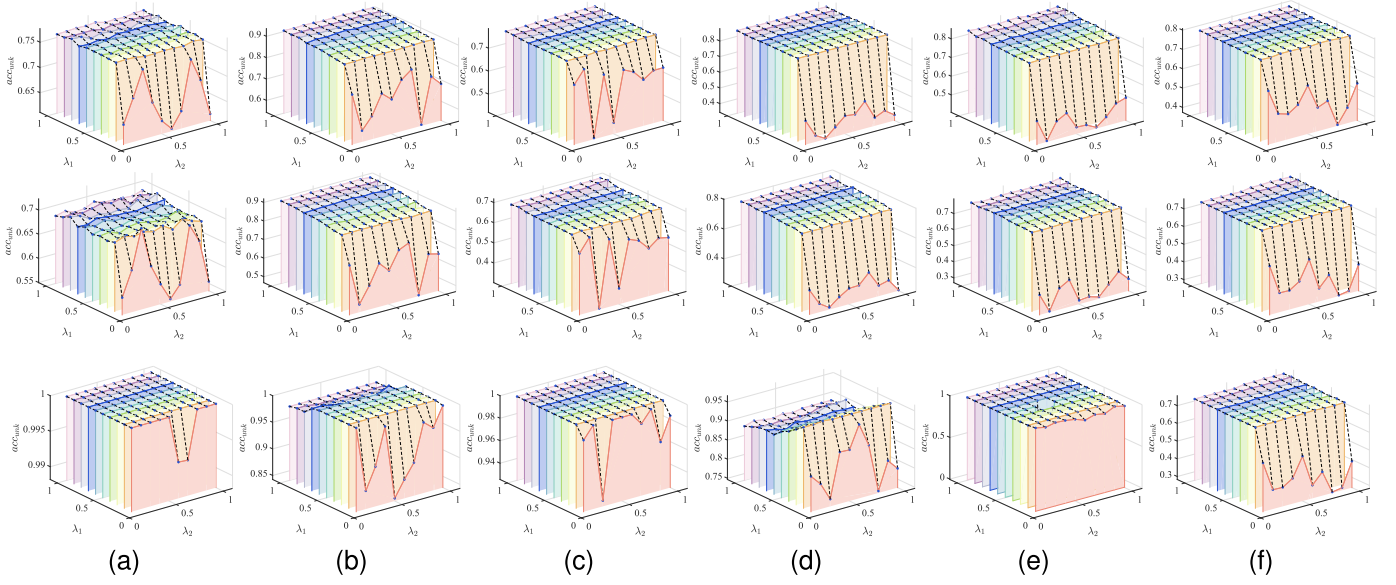


Fig. 9. Impact of hyperparameters λ_1 and λ_2 on acc_{Total} , acc_{com} , and acc_{pri} on the six scenarios of CWRU dataset.

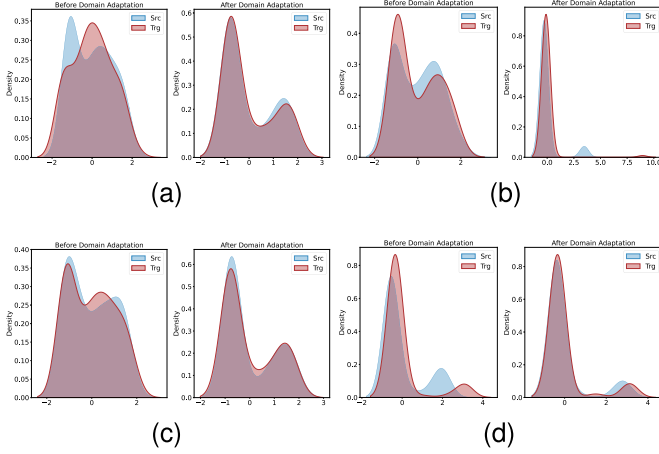


Fig. 10. Feature distribution of common samples from source and target domains on CWRU dataset. (a) 1-to-0 class 5. (b) 1-to-0 class 6. (c) 2-to-1 class 5. (d) 3-to-2 class 4.

4) *Hyperparameter Selection*: To investigate the sensitivity of hyperparameters λ_1 and λ_2 , a grid search is conducted on the CWRU dataset, varying both parameters within the range of $\{0.1, 0.2, \dots, 1.0\}$ across the six scenarios, as illustrated in Fig. 9. The observations indicate that the model performance stabilizes and remains consistently high when $\lambda_2 \geq 0.2$. Specifically, while acc_{pri} exhibits a slight variation—peaking at $\lambda_2 = 0.2$ and marginally decreasing afterward due to the complexity of private classes—it maintains a high standard ($\approx 90\%$). Critically, the interaction between the two parameters reveals that effective local alignment (λ_2) is contingent upon the solid foundation provided by global alignment (λ_1). Once λ_1 ensures sufficient global matching, increasing λ_2 further refines the boundaries. Overall, TSFA demonstrates strong robustness, achieving optimal performance across a broad parameter space without the need for extensive fine-tuning.

5) *Distribution Shift*: Fig. 10 visualizes the feature distributions of common classes across domains using kernel density estimation (KDE) [20]. It can be observed that TSFA

TABLE X
SHADED AREA OF THE PROBABILITY DENSITY CURVE ON THE CWRU DATASET

scenarios	TSFA	source area	target area	overlap	overlap ratio
Fig. 8 (a)	✗	0.5553	0.5244	0.5242	0.9440
	✓	0.5211	0.5144	0.5060	0.9710
Fig. 8 (b)	✗	0.7974	0.7980	0.7674	0.9624
	✓	0.6207	0.6353	0.6159	0.9923
Fig. 8 (c)	✗	0.8141	0.7253	0.7402	0.9092
	✓	0.5135	0.4999	0.5060	0.9854
Fig. 8 (d)	✗	0.8480	0.8949	0.8085	0.9534
	✓	0.6353	0.6266	0.6342	0.9983

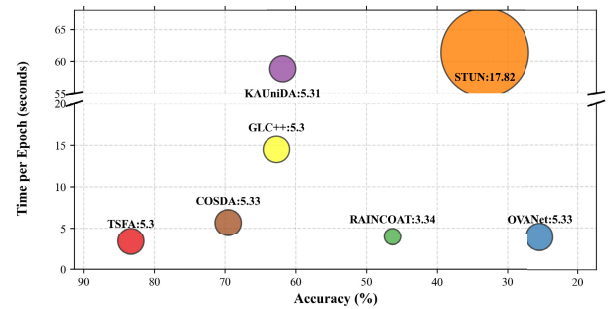


Fig. 11. Training performance versus time per epoch. The size of each bubble indicates the model parameter count (10^5).

effectively transforms the originally disjoint source and target distributions into consistent shapes with aligned peaks. Furthermore, the quantitative results in Table X show a marked increase in the distributional overlap ratio. These results demonstrate that TSFA successfully aligns feature distributions and reduces the domain gap.

6) *Computational Efficiency*: Fig. 11 illustrates the computational efficiency on the CWRU dataset ($3 \rightarrow 0$). With 3.38 s per epoch, TSFA is significantly faster than clustering-based methods (e.g., GLC++ at 14.49 s) due to its optimized

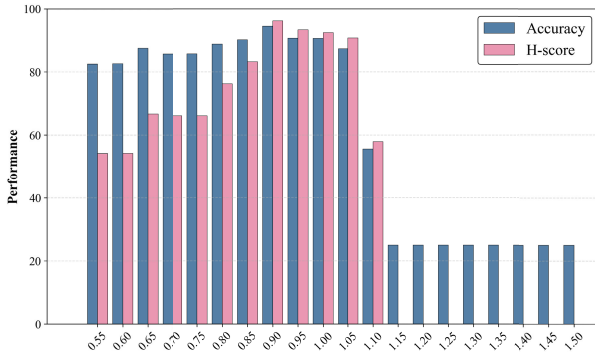


Fig. 12. Acc and H-score under varying threshold in HAR dataset.

matrix operations. With 0.53×10^5 parameters, TSFA remains lightweight compared to heavy architectures, like STUN (1.78×10^5).

7) *Sensitivity to Threshold*: Fig. 12 presents a sensitivity analysis of the rejection threshold $th = \lambda_3/C^s$, where λ_3 is swept across the range [0.5, 1.5] with a step size of 0.05. The results demonstrate that the final accuracy is stable around the theoretical anchor $\lambda_3 = 1.0$. These observations corroborate the theoretical premise that $1/C^s$ functions as an effective boundary for uncertainty rejection.

V. CONCLUSION

In this article, a TSFA method for UOSDA in time-series classification is proposed, which focuses on addressing the challenges of nonstationarity and the risk of negative transfer. To be specific, a time–frequency feature extractor is utilized to capture domain-invariant representations across diverse operating conditions. A TSFA strategy is proposed to mitigate the negative transfer caused by gradient instability resulting from large interclass variations in frequency features, as well as the lack of classwise alignment. In the global alignment stage, cosine similarity, which provides more stable gradients, is used alongside an SDE to implicitly cluster all target-domain samples in the feature space, achieving a coarse alignment. Subsequently, a distribution-aware pseudolabeling method, OSAM, is designed to perform precise classwise alignment. Furthermore, by leveraging an adaptive decision boundary, OSAM effectively identifies target-private samples. Extensive experiments on three benchmark datasets demonstrated the superiority of TSFA.

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